

Electronic supplementary material

Units matter: cooperation in large language model agents tracks how the payoff matrix is written, not what it means

This document holds the per-model and per-cell breakdowns behind the summary statistics of the main text, and the validation of the strategy read-out. Every table is generated directly from the machine-readable analysis output by `Analysis/scripts/42_supp_tables.py`; none of the numbers is transcribed by hand, and re-running the pipeline regenerates this document and the main text together.

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S1 Validating the strategy read-out

The read-out described in §2.4 of the main text assigns each agent-game a label from the four canonical memory-one rules in two stages: an enumeration that proves whether a trajectory is exactly what a rule would have played, and where no rule fits, a long short-term memory network [1] that returns the nearest one. This section records what each stage is worth, because the strategy results of §3.8 rest on it and because two thirds of the labels come from the learned stage.

Validation corpus. The network is trained and evaluated on synthetic play generated from the same four rules, at execution-noise levels of 0 and 5 per cent, with a separate network for each of the two horizons in the corpus. Ten per cent and twenty per cent noise are held out entirely, and are reported below because real transcripts are noisier than the training distribution and it matters how the network degrades outside it.

Accuracy. At ten rounds the pipeline recovers the generating rule with accuracy 0.995 on noise-free play and 0.975 at five per cent noise, against Bayes-optimal single-label ceilings of 0.996 and 0.987. At thirty rounds the corresponding figures are 0.953 at five per cent noise against a ceiling of 0.966. Where exactly one rule fires the label is right 99.5 per cent of the time at ten rounds and 98.7 per cent at thirty, which is what makes the provable stage worth running first even though it covers a minority of games.

The identifiability floor. On noise-free thirty-round play the forced single-label ceiling is 0.625, and this is a property of the game rather than a failure of the method: an agent that cooperates throughout against a cooperator is exactly consistent with AllC, tit-for-tat and win-stay lose-shift at once, and no procedure can separate them. Allowing the first stage to return the *set* of rules that fit, the answer is correct on 100 per cent of that file with a mean answer size of 2.13 labels. This is why the main text records such runs as *ambiguous* rather than resolving them, and why the ambiguous share is treated throughout as a statement about identifiability rather than about strategy.

Where the corpus sits relative to the training distribution. The exact-rule coverage the read-out achieves on the real corpus locates it against the synthetic noise levels. In the open-weight arm 22.1 per cent of agent-games carry a provable rule, against 21.8 per cent for thirty-round synthetic play at five per cent noise, so that arm sits almost exactly at the validated operating point. In the frontier arm the figure is 25.5 per cent, below the 39.1 per cent of ten-round play at ten per cent noise and above the 14.3 per cent at twenty per cent noise. Frontier transcripts therefore depart from the canonical rules by more than the training corpus does, and the network is extrapolating there more often. On those two held-out noise levels its accuracy where no rule fits is 0.908 and 0.788, which brackets what the frontier labels are worth and is the reason §3.8 reports every test on the four named rules as well as on all five categories. Table S7 gives the full provenance breakdown, scale by scale.

S2 Supplementary tables

Table S1: The corpus, by model. “Cell n ” is the number of agent-games in every model $\times$ language $\times$ scale cell; the design is balanced, with none missing. The two arms are never pooled.

Arm	Model	Scales	Range	Lang.	Rnds	Dyads	Games	Decis.	Cell n	Coop.
open-weight	Gemma 3 12B	6	0.01–1000	5	30	1,200	2,400	72,000	80	0.535
open-weight	Llama 3.1 8B	6	0.01–1000	5	30	1,200	2,400	72,000	80	0.420
open-weight	Qwen3 8B	6	0.01–1000	5	30	1,200	2,400	72,000	80	0.474
frontier	Claude 3.5 Haiku	3	0.1–10	5	10	600	1,200	12,000	80	0.625
frontier	GPT-4o	3	0.1–10	5	10	600	1,200	12,000	80	0.448
frontier	Gemini 3.5 Flash-Lite	3	0.1–10	5	10	600	1,200	12,000	80	0.740
frontier	Mistral Large	3	0.1–10	5	10	600	1,200	12,000	80	0.524

Table S2: What the rescaling changes, and what it cannot. The cell values as the prompt template prints them at each scale, with the notation features derived from that string on the left and the game-theoretic invariants on the right. Greed, fear, the dominant action and the equilibrium are constant down each arm by construction; only the printed string moves.

Arm	λ	Printed cells	Regime	Frac.	Glyphs	Digits	Greed	Fear	Dom.	Nash
open-weight	0.01	0 / 0.02 / 0.08 / 0.1	fractional	1	3.00	1	0.20	0.20	A	A,A
open-weight	0.1	0 / 0.2 / 0.8 / 1	fractional	1	2.00	1	0.20	0.20	A	A,A
open-weight	1	0 / 2 / 8 / 10	unit	0	1.25	2	0.20	0.20	A	A,A
open-weight	10	0 / 20 / 80 / 100	unit	0	2.00	3	0.20	0.20	A	A,A
open-weight	100	0 / 200 / 800 / 1000	large	0	2.75	4	0.20	0.20	A	A,A
open-weight	1000	0 / 2000 / 8000 / 10000	large	0	3.50	5	0.20	0.20	A	A,A
frontier	0.1	0 / 0.2 / 0.6 / 1	fractional	1	2.00	1	0.20	0.40	A	A,A
frontier	1	0 / 2 / 6 / 10	unit	0	1.25	2	0.20	0.40	A	A,A
frontier	10	0 / 20 / 60 / 100	unit	0	2.00	3	0.20	0.40	A	A,A

Table S3: The description ladder, model by model. Δ BIC from the best specification for that model, every row fitted to identical data with language and both personas controlled. Bold marks the best specification in each column. Five of the seven models prefer a description built from how the cells are printed; Gemma 3 12B prefers a cubic in $\log_{10} \lambda$, the one case in the corpus where the notation account loses to a polynomial, and Mistral Large prefers no λ term at all, which is the correct description of a model whose cooperation rate did not move. Over three scales a cubic is collinear with a quadratic and the fractional flag is the same specification as the regime factor.

How λ enters	open-weight, 6 scales			frontier, 3 scales			
	Gemma	Llama	Qwen3	Claude	GPT-4o	Gemini	Mistral
no λ term	102.1	687.7	292.1	243.3	118.4	7.4	0.0
linear in $\log_{10} \lambda$	71.5	392.0	237.2	61.3	29.3	6.5	5.7
quadratic in $\log_{10} \lambda$	19.2	257.6	149.2	6.9	7.0	6.2	11.9
cubic in $\log_{10} \lambda$	0.0	225.2	146.4	6.9	8.1	6.2	11.9
fractional flag	34.1	33.5	75.9	0.0	0.0	0.0	4.9
fractional + glyph count	9.2	31.8	51.2	6.9	7.0	6.2	11.9
notation regime	24.3	0.0	0.0	0.0	0.0	0.0	4.9
saturated in λ	3.3	1.0	11.6	6.9	7.0	6.2	11.9

Table S4: The persona instruction, model by model and scale by scale. The effect is the cooperation rate of an agent told to be cooperative minus that of an agent told to be selfish. A dagger marks an estimate whose 95% dyad bootstrap interval excludes zero. Read across a row: five of the seven models hold one sign over the whole sweep, which is why the average over models, which does cross zero, describes no individual model.

Arm	Model	$\lambda = 0.01$	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$	$\lambda = 100$	$\lambda = 1000$
open-weight	Gemma 3 12B	+0.037	+0.239 [†]	+0.396 [†]	+0.368 [†]	+0.152 [†]	+0.201 [†]
open-weight	Llama 3.1 8B	+0.115 [†]	+0.072 [†]	+0.140 [†]	+0.151 [†]	+0.101 [†]	+0.093 [†]
open-weight	Qwen3 8B	-0.437 [†]	-0.448 [†]	-0.322 [†]	-0.179 [†]	-0.314 [†]	-0.302 [†]
frontier	Claude 3.5 Haiku	-	-0.318 [†]	-0.067 [†]	-0.057 [†]	-	-
frontier	GPT-4o	-	-0.438 [†]	+0.200 [†]	+0.305 [†]	-	-
frontier	Gemini 3.5 Flash-Lite	-	-0.489 [†]	+0.238 [†]	+0.198 [†]	-	-
frontier	Mistral Large	-	-0.599 [†]	-0.255 [†]	-0.152 [†]	-	-

Table S5: Strategy-mix shifts on both vocabularies. The largest total-variation distance between the mixes at two payoff scales, against a null that shuffles the scale label across whole dyads. “5-cat.” uses the five categories the read-out emits; “named” drops *ambiguous* and renormalises over the four canonical rules, so that a change in how identifiable the play is cannot contribute. “Named share” is the fraction of that model’s agent-games the restriction keeps. Every model is significant on both vocabularies. Gemma 3 12B is the one whose estimate changes materially, because almost half of its five-category distance is movement into the ambiguous bucket.

Arm	Model	5-cat.	P	null	named	P	null	Named share
open-weight	pooled	0.254	< 0.001	0.075	0.226	< 0.001	0.072	0.882
open-weight	Gemma 3 12B	0.375	< 0.001	0.142	0.205	< 0.001	0.130	0.686
open-weight	Llama 3.1 8B	0.387	< 0.001	0.115	0.387	< 0.001	0.115	1.000
open-weight	Qwen3 8B	0.278	< 0.001	0.122	0.275	< 0.001	0.119	0.961
frontier	pooled	0.206	< 0.001	0.056	0.222	< 0.001	0.051	0.917
frontier	Claude 3.5 Haiku	0.262	< 0.001	0.115	0.245	< 0.001	0.108	0.953
frontier	GPT-4o	0.325	< 0.001	0.102	0.296	< 0.001	0.098	0.942
frontier	Gemini 3.5 Flash-Lite	0.338	< 0.001	0.120	0.304	< 0.001	0.110	0.783
frontier	Mistral Large	0.125	0.0055	0.097	0.127	0.0040	0.094	0.988

Table S6: The strategy mix, model by model. Share of that model’s agent-games carrying each label at each payoff scale. The frontier arm sweeps only $\lambda \in \{0.1, 1, 10\}$.

Model	Label	$\lambda = 0.01$	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$	$\lambda = 100$	$\lambda = 1000$
Gemma 3 12B	AllC	0.160	0.165	0.170	0.180	0.100	0.142
	TFT	0.328	0.240	0.113	0.188	0.128	0.140
	WSLS	0.025	0.045	0.010	0.007	0.022	0.050
	AllD	0.393	0.290	0.247	0.290	0.370	0.312
	Ambiguous	0.095	0.260	0.460	0.335	0.380	0.355
Llama 3.1 8B	AllC	0.172	0.138	0.375	0.427	0.347	0.323
	TFT	0.090	0.065	0.113	0.090	0.110	0.113
	WSLS	0.075	0.030	0.133	0.090	0.085	0.107
	AllD	0.662	0.767	0.380	0.393	0.458	0.458
	Ambiguous	0.000	0.000	0.000	0.000	0.000	0.000
Qwen3 8B	AllC	0.315	0.278	0.460	0.455	0.390	0.388
	TFT	0.030	0.035	0.075	0.087	0.090	0.095
	WSLS	0.048	0.077	0.055	0.100	0.052	0.050
	AllD	0.537	0.590	0.350	0.312	0.458	0.438
	Ambiguous	0.070	0.020	0.060	0.045	0.010	0.030
Claude 3.5 Haiku	AllC	-	0.282	0.407	0.438	-	-
	TFT	-	0.207	0.193	0.233	-	-
	WSLS	-	0.145	0.217	0.175	-	-
	AllD	-	0.350	0.102	0.110	-	-
	Ambiguous	-	0.015	0.080	0.045	-	-
GPT-4o	AllC	-	0.225	0.290	0.255	-	-
	TFT	-	0.087	0.133	0.138	-	-
	WSLS	-	0.115	0.250	0.250	-	-
	AllD	-	0.568	0.273	0.242	-	-
	Ambiguous	-	0.005	0.055	0.115	-	-
Gemini 3.5 Flash-Lite	AllC	-	0.307	0.412	0.388	-	-
	TFT	-	0.072	0.180	0.145	-	-
	WSLS	-	0.098	0.223	0.198	-	-
	AllD	-	0.233	0.020	0.075	-	-
	Ambiguous	-	0.290	0.165	0.195	-	-
Mistral Large	AllC	-	0.398	0.460	0.425	-	-
	TFT	-	0.020	0.083	0.048	-	-
	WSLS	-	0.165	0.098	0.168	-	-
	AllD	-	0.403	0.345	0.355	-	-
	Ambiguous	-	0.015	0.015	0.005	-	-

Table S7: Where each strategy label came from. Share of agent-games resolved at each stage of the read-out. “Provable rule” means exactly one canonical rule reproduces the run of play; “several rules” means more than one does and the run is recorded as ambiguous; the last two are the LSTM’s nearest rule, split at the 0.90 posterior floor below which a read-out designed to abstain would decline to answer. None of the four shares is constant across the sweep, so how nameable the play is depends on the payoff scale.

Arm	Scale	provable rule	several rules	LSTM ≥ 0.90	LSTM < 0.90
open-weight	$\lambda = 0.01$	0.284	0.055	0.497	0.163
open-weight	$\lambda = 0.1$	0.255	0.093	0.517	0.135
open-weight	$\lambda = 1$	0.193	0.173	0.465	0.169
open-weight	$\lambda = 10$	0.172	0.127	0.487	0.215
open-weight	$\lambda = 100$	0.218	0.130	0.497	0.154
open-weight	$\lambda = 1000$	0.207	0.128	0.512	0.153
open-weight	all scales	0.221	0.118	0.496	0.165
frontier	$\lambda = 0.1$	0.360	0.081	0.339	0.219
frontier	$\lambda = 1$	0.215	0.079	0.444	0.262
frontier	$\lambda = 10$	0.190	0.090	0.430	0.290
frontier	all scales	0.255	0.083	0.405	0.257

Table S8: Cooperation by prompt language and payoff scale. The FAIRGAME templates do not state the objective identically in every language, so language is confounded with objective framing and the framing each template uses is given alongside it. The five languages do not move in parallel across the sweep.

Arm	Language	Framing	$\lambda = 0.01$	$\lambda = 0.1$	$\lambda = 1$	$\lambda = 10$	$\lambda = 100$	$\lambda = 1000$
open-weight	English	minimise	0.314	0.380	0.697	0.673	0.449	0.474
open-weight	French	minimise	0.365	0.460	0.608	0.606	0.554	0.603
open-weight	Vietnamese	minimise	0.501	0.443	0.615	0.565	0.586	0.659
open-weight	Chinese	maximise	0.345	0.336	0.529	0.566	0.539	0.541
open-weight	Arabic	maximise	0.260	0.299	0.413	0.423	0.252	0.232
frontier	English	minimise	-	0.483	0.627	0.614	-	-
frontier	French	minimise	-	0.438	0.693	0.673	-	-
frontier	Vietnamese	minimise	-	0.474	0.604	0.628	-	-
frontier	Chinese	maximise	-	0.517	0.592	0.608	-	-
frontier	Arabic	maximise	-	0.534	0.650	0.629	-	-

Table S9: What one decade of separation buys. The cooperation gap between two adjacent scales, held against the 95th percentile of the permutation null for that model’s *full* sweep, which is conservative because a two-level design generates a smaller null. This is the evidence behind the reporting standard: a single extra condition one decade away detects the effect in six of the seven models.

Arm	Model	Contrast	Gap	Null (full sweep)	Clears
open-weight	Gemma 3 12B	0.01 vs 0.1	0.171	0.121	yes
open-weight	Gemma 3 12B	0.1 vs 1	0.145	0.121	yes
open-weight	Gemma 3 12B	1 vs 10	0.061	0.121	no
open-weight	Gemma 3 12B	10 vs 100	0.088	0.121	no
open-weight	Gemma 3 12B	100 vs 1000	0.064	0.121	no
open-weight	Llama 3.1 8B	0.01 vs 0.1	0.038	0.047	no
open-weight	Llama 3.1 8B	0.1 vs 1	0.182	0.047	yes
open-weight	Llama 3.1 8B	1 vs 10	0.016	0.047	no
open-weight	Llama 3.1 8B	10 vs 100	0.048	0.047	yes
open-weight	Llama 3.1 8B	100 vs 1000	0.000	0.047	no
open-weight	Qwen3 8B	0.01 vs 0.1	0.053	0.095	no
open-weight	Qwen3 8B	0.1 vs 1	0.240	0.095	yes
open-weight	Qwen3 8B	1 vs 10	0.027	0.095	no
open-weight	Qwen3 8B	10 vs 100	0.136	0.095	yes
open-weight	Qwen3 8B	100 vs 1000	0.014	0.095	no
frontier	Claude 3.5 Haiku	0.1 vs 1	0.232	0.058	yes
frontier	Claude 3.5 Haiku	1 vs 10	0.007	0.058	no
frontier	GPT-4o	0.1 vs 1	0.226	0.062	yes
frontier	GPT-4o	1 vs 10	0.008	0.062	no
frontier	Gemini 3.5 Flash-Lite	0.1 vs 1	0.080	0.062	yes
frontier	Gemini 3.5 Flash-Lite	1 vs 10	0.020	0.062	no
frontier	Mistral Large	0.1 vs 1	0.036	0.065	no
frontier	Mistral Large	1 vs 10	0.005	0.065	no

Table S10: Cooperation across the game, by notation regime. Round one on its own, then three equal blocks of the remaining rounds. The gap between the regimes is open on round one, on which no history exists. The two arms divide different horizons into the same four positions and are not comparable to each other.

Arm	Regime	r1	r2-5	r6-15	r16-30
open-weight	fractional	0.366	0.314	0.358	0.394
open-weight	unit	0.495	0.500	0.554	0.604
open-weight	large	0.445	0.407	0.473	0.524
frontier	fractional	0.477	0.427	0.482	0.529
frontier	unit	0.676	0.630	0.629	0.624

References

- [1] Sepp Hochreiter and Jürgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8):1735–1780, 1997. doi: 10.1162/neco.1997.9.8.1735.